NASA SPACE APPS CHALLENGE 2026

AstroCare

AI Health Sentinel for Astronauts from Launch to Recovery

by **Team AstroZen**

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health modules, launch to recovery	machine-learning models (M1, M3)	rule engines against each person's own baseline	diagnoses: every flag shows its evidence

“Nothing is diagnosed. Everything is evidenced.”

Challenge: Create Health Monitoring Software for Astronauts on Space Missions
Data: NASA OSDR · NASA DONKI · NOAA GOES · MSL/RAD · published studies

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1. Project Title and Summary

AstroCare: An AI Health Sentinel for Astronauts, from Launch to Recovery by Team AstroZen.

AstroCare is an evidence-first crew health console. It brings space weather, radiation dose, blood radiation response, immune and metabolic lab panels, microbiome, telomere evidence, daily vital signs and self-reports into one place, and compares every astronaut with their own normal rather than a population average. Only two parts are true machine-learning models, because only they have enough suitable NASA data to train and validate honestly: **M1** forecasts solar particle storms and **M3** reads radiation dose from blood gene expression. Everything else measures change against each person's own pre-flight baseline with fixed, transparent rules, estimates dose from physics, or shows published context. Every value on screen carries its source, its rule and a safe, non-diagnostic next step.

Table 1. *Project at a glance*

Item	AstroCare
Problem	Astronauts face health risks they cannot feel: radiation, microgravity, isolation. Their data sit in separate systems, and on deep-space missions a doctor on Earth can be up to twenty minutes away by message.
Solution	One console that warns early where data support a prediction, measures change against each crew member's own baseline everywhere else, and explains every flag.
Primary users	Astronauts and crew (on-board self-monitoring)
Secondary users	Flight surgeons and mission medical operations (review of flags and evidence)
Tertiary users	Researchers (provenance-labelled data, methods and code)
Output language	"Changed from baseline", "Monitoring recommended", "Back within own variation". Never a disease name, a medication or a single overall health score.
Runs	Fully offline (no internet at run time): Python backend + browser console, one command (python run.py)

2. Project Objectives

The overall objective is to give a crew member, and the flight surgeon who supports them, a single, trustworthy picture of crew health during and after a mission: what changed, when, how the system knows, and what to do next. The specific objectives are:

1. **Warn early about radiation storms** using a calibrated machine-learning forecast trained on NASA solar-event history (M1).
2. **Estimate crew radiation dose** behind different shielding from measured cosmic-ray rates (M2).
3. **Read biological radiation response from blood** with machine-learning biodosimetry models that refuse to answer outside their validated time and dose windows (M3).
4. **Monitor body systems against each person's own baseline**: microbiome (M4), immune and inflammatory markers (M7), recovery after landing (M6) and daily vital signs (M9).
5. **Show published evidence where data are too thin for numbers**: brain (M5), telomeres (M8) and bone.
6. **Explain every flag and recommend a safe next step** through an explanation layer and a rule-based Action Center that never diagnoses.
7. **Label the provenance of every value** (measured, derived, ML-predicted, estimated, context, self-reported, telemetry) so predictions and measurements can never be confused.
8. **Work without Earth**: all models, rules and data run locally.

2.1 Features of the system

Table 2. *All features, with the kind of analysis behind each*

#	Feature	Module	Method type
1	Solar particle storm early warning (48 h)	M1	Machine learning (calibrated logistic regression)
2	Crew radiation dose by location (habitat, shelter, EVA)	M2	Physics estimate
3	Blood radiation biodosimeter: dose at 20–28 h; dose class at 1, 2, 6 h	M3	Machine learning (elastic-net)
4	Microbiome drift from own pre-flight samples	M4	Rule vs own baseline (M4-DRIFT-v1)
5	Brain-risk evidence card	M5	Published context (animal studies)
6	Recovery after landing (R+1 to R+194)	M6	Rule vs own baseline (M6-TRAJ-v2)
7	Immune and inflammatory monitoring (serum panels, blood counts)	M7	Rule vs own baseline (M7-DEV-v1)
8	Telomere evidence card	M8	Published context
9	Crew Health Console: daily vitals, live heart rate, flag feed	M9	Rules vs own 30-day baseline (M9-FLAG-v1, M9-MULTI-v1, M9-Q01)
10	Mission context: phase, gravity, isolation, cabin CO2	Context	Context only
11	Bone panel: change from own pre-flight DXA scan	Bone	Measured + derived; published context
12	Daily self-check (sleep, stress, mood, fatigue) and reaction-time test	Behaviour	Self-report + in-app measurement, rule R50

#	Feature	Module	Method type
13	Explanation card for every flag	All	What changed · when · rule · evidence
14	Action / Recommendation Center	All	Rules R02–R99 in priority order
15	Provenance record and evidence registry for every value	All	7 status types; 14 evidence entries

Machine learning vs rules: the evidence gate

A domain gets a machine-learning model only when suitable human (or human-cell) data, a real target and enough independent subjects exist. That is true for solar storms (decades of events) and for blood biodosimetry (ex vivo irradiated human blood from 5–11 donors per model).

The Inspiration4 crew data cover four people, which is too few to train and validate a model honestly. Those domains are monitored with deterministic rules against each person's own pre-flight baseline, an approach that stays valid even for a single person.

3. Motivation

Space is hard on the human body in ways the body does not announce. Ionising radiation from galactic cosmic rays and solar particle events damages DNA without any sensation; microgravity shifts fluids, weakens bone and changes the heart; isolation and confinement affect sleep, mood and reaction time. On the International Space Station a flight surgeon can talk to the crew in real time. On the Moon the delay is seconds, and on the way to Mars a message can take up to twenty minutes each way. Crews on future missions will need to check their own health and act sensibly before help can answer.

Health data from spaceflight are also scattered. Radiation forecasts live in space-weather systems, dosimetry in instrument archives, blood and microbiome results in research repositories, and wearable data on devices. NASA's Open Science Data Repository (OSDR) now hosts the SpaceX Inspiration4 mission data, the first openly available biological data from an all-civilian crew, but those data are measured for research, not presented for the person they came from.

Finally, much health software either predicts without enough data or gives a single score that hides the evidence. Four people cannot support a population model; a black-box score cannot be checked by a doctor twenty light-minutes away. We wanted the opposite: a system that **predicts only where NASA data support a prediction**, **compares each astronaut with their own normal** everywhere else, and **always shows the rule and the source** behind what it says. That is the motivation for AstroCare.

> **Design principles**
>
> 1. Compare each astronaut with their own normal, not a population threshold.
> 2. Warn early only where NASA data support a prediction; otherwise measure, show context, or ask the astronaut.
> 3. Never diagnose, never name a medication, never give one overall health score.
> 4. Show the evidence for every flag: what changed, when, which rule fired, and which study supports it.
> 5. Refuse to answer outside a model's validated window.
> 6. Run without Earth: no internet at run time.

4. Introduction: What We Built and How

AstroCare is a complete software system made of four layers: a **data layer** of NASA and published datasets stored with the software, an **analysis layer** of two machine-learning models, one physics estimate and six rule engines, an **API layer** that turns every result into a provenance-labelled indicator, and a **12-page web console** for the crew and the flight surgeon. Figure 1 shows how a value travels from source data to a recommendation.

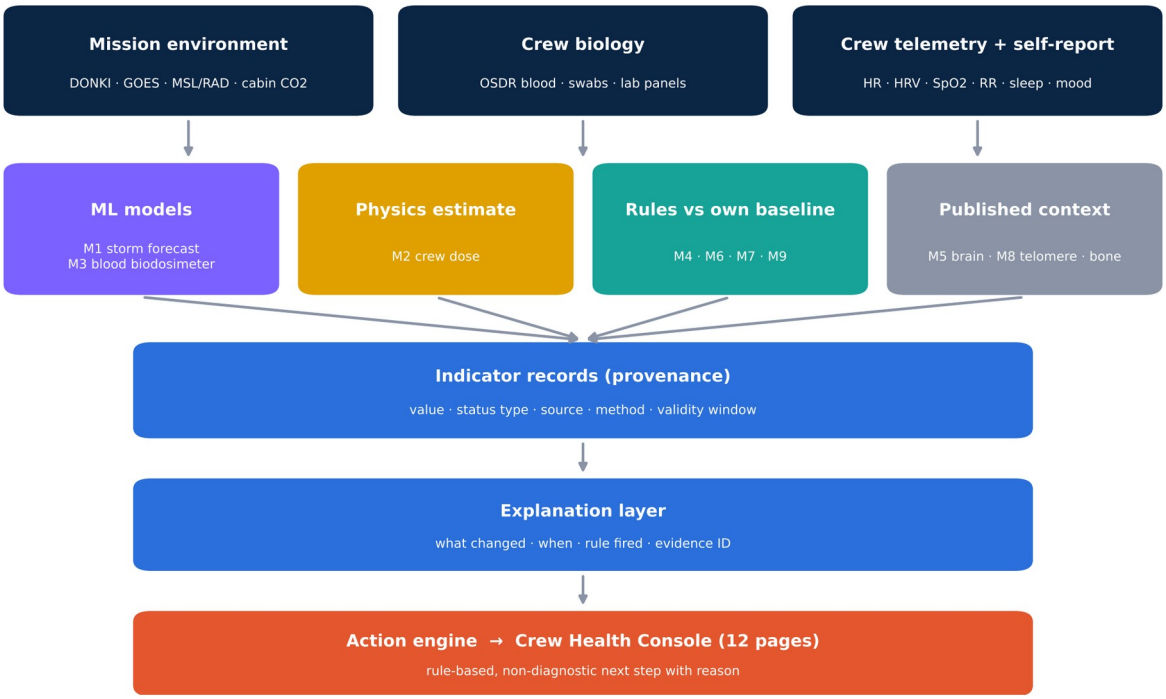


Figure 1. *AstroCare architecture: every value travels from data to an indicator record, an explanation and a rule-based action.*

4.1 How it was built

Table 3. *Technology stack*

Layer	Technology	What it does
Analysis notebooks	Python, Google Colab (pandas, scikit-learn, limma/oligo where needed)	Data cleaning, model training and validation for M1, M3, M4, M9
Backend	Python 3.9+; FastAPI + uvicorn (optional); built-in standard-library server as fallback	33 API endpoints; models and rules run in pure Python (no scikit-learn needed at run time)
Rule set	Versioned JSON (rules_v1.json), evidence registry (14 entries)	All thresholds, rule texts and evidence links in one place
Frontend	HTML, CSS, JavaScript; canvas charts; Three.js 3D hero scene	12 pages: Home, Console, Radiation, Blood, Microbiome, Immune, Telomere, Recovery, Body & Mind, Actions, Evidence, About
Testing	Python unittest (26 tests) + headless-browser checks	Reproduction of notebook results, safety wording, provenance, every page and endpoint

4.2 Two crews, never mixed

The console works with two kinds of subject. The **mission crew (SIM-A, SIM-B, SIM-C, SIM-D)** is a four-person deep-space mission scenario running from mission day −45 to +45; it drives the Console (M9 vitals), the Radiation page (M1/M2 for the current mission), the Blood page (M3 on the crew's blood profiles) and Body & Mind. The **Inspiration4 crew (C001–C004)** are the de-identified NASA OSDR subjects; they drive Microbiome (M4), Immune (M7), Telomere (M8) and Recovery (M6). The crew selector switches automatically with the page, so measurements from one group are never shown under the other's name.

5. Data

This chapter lists every dataset used, from the first download to the form the software uses. Raw files were never edited: each module has its own cleaned tables, and every displayed value can be traced back to a file and a processing step.

Table 4. *All datasets used by AstroCare*

Module	Dataset	Subjects / size	What it contains	Used as
M1	NASA DONKI flare + SEP catalogue; NOAA GOES ≥10 MeV proton flux	Flares 2010–2026; train 1,167 flares (39 storms), test 1,720 forecast points (21 storms)	Flare class, location, timing; particle events; proton background	Training and test data
M2	MSL/RAD (Curiosity cruise to Mars), Zeitlin et al. 2013	Mission-long dose rate	GCR dose-equivalent ≈ 1.8 mSv/day behind spacecraft shielding	Physical parameter
M3	OSDR OSD-1012 (GEO GSE90909)	12 donors (11 usable); X-ray 0–4 Gy and neutrons 0.25–1 Gy; 24 h	Agilent microarrays, ex vivo human blood	Training (24 h) + neutron test
M3	OSDR OSD-410	10 donors; 0 / 0.15 / 0.30 / 1.5 Gy γ; 1, 2, 6 h	Affymetrix PrimeView arrays	Training (early models)
M3	OSDR OSD-71	4 donors; ~1 h; different lab and platform	Affymetrix HuGene 1.0 ST arrays	External test only
M4	OSDR OSD-572 (Inspiration4 swabs)	4 crew, 97 samples, 8 timepoints incl. FD2, FD3	Skin and oral swab metagenomics	Rule input
M6, M7	OSDR OSD-575 serum: Eve 71-plex, Alamar NULISA 203-target, Eve cardiovascular, CMP	4 crew × 7 timepoints (L-92 … R+194)	Cytokines, inflammation proteins, metabolic panel	Rule input
M6, M7	OSDR OSD-569 whole blood: CBC	4 crew × 7 timepoints	20 blood-count analytes	Rule input
M8	Garcia-Medina et al. 2024 (PMC11022651)	4 crew; dried blood spots	Telomere length by qPCR (published summary)	Context
M5	OSDR OSD-611, OSD-374, OSD-479	Mice	Hippocampus RNA-seq and behaviour after space-type radiation	Context
M9	Mission crew telemetry SIM-A … SIM-D	4 crew × 4 vitals × 90 mission days	Resting HR, HRV (RMSSD), SpO2, respiratory rate; cabin CO2	Rule input
Bone	NASA skeletal recovery study (NTRS 20070001978)	Long-duration crew	Recovery of lost bone after landing	Context
Behaviour	In-app entries	Per crew member	Sleep, stress, mood, fatigue; reaction time	Self-report / measured

5.1 M1: space weather (NASA DONKI + NOAA GOES)

Flare records (class, peak time, source longitude and latitude, rise time) were taken from the NASA DONKI catalogue, and solar energetic particle (SEP) storms were defined from NOAA GOES ≥10 MeV proton flux crossing 10 pfu (an S1 storm).

- **Target:** the parent flare of a NOAA GOES ≥10 MeV ≥10 pfu SEP within 48 h; only the largest flare per SEP onset is labelled positive, so one storm is counted once.
- **Forecast points:** each flare gives a forecast at its peak (1,367 test points) and, when a coronal mass ejection follows, a second point at flare + CME (353 test points).
- **Features prepared:** flare_log10_peak_flux, flare_lon_deg, flare_lat_deg, flare_rise_min, log10_bg_max_24h_pfu, n_eligible_flares_prior_24h, n_noaa_sep_onsets_prior_72h, bg_coverage_24h, low_coverage. Every feature uses only data available before the forecast time.
- **Split by time:** train 2010-06-12 to 2023-12-24 (1,167 flares, 39 storms); test 2024-01-01 to 2026-10-19 (1,720 forecast points, 21 storms).
- **Excluded:** indeterminate labels; flares with no GOES background (28 Oct 2021); non-parent flares of the same SEP; CME features (unreliable linkage, leakage risk).

5.2 M2: cosmic-ray dose (MSL/RAD)

The Radiation Assessment Detector on the Curiosity rover measured a galactic cosmic-ray dose-equivalent rate of about 1.8 mSv/day inside the spacecraft during the cruise to Mars (Zeitlin et al. 2013). M2 uses this as its background rate. The storm term uses the ≥10 MeV proton flux profile of the mission's solar particle event (onset on mission day 33, peak 2400 pfu) and shielding factors for three locations (habitat 1.0, storm shelter 0.35, EVA 3.5).

5.3 M3: blood gene expression (OSD-1012, OSD-410, OSD-71)

Three NASA OSDR studies of human blood irradiated outside the body (ex vivo) were used. The studies were **never pooled**: each has its own platform, time point and radiation type, and each model is trained on one study only.

- **Version 1** was built from the GEO series matrices (GSE90909, GSE63952, GSE64375).
- **Version 2 (used in the software)** was rebuilt from one cleaned long-format file, m3_clean_ml_dataset.csv: 6,602,840 rows, 208 samples, 3 studies.
- **Quality filters:** rows flagged MISSING_IN_SOURCE, MISSING_SOME_SAMPLES, CONTROL_PROBE, UNMAPPED or AMBIGUOUS_MULTI_GENE were dropped, leaving 5,484,176 rows. Genes kept: OSD-410 46,869; OSD-71 21,235; OSD-1012 4,973 (pre-filtered by the submitters).
- **Own-baseline subtraction:** every exposed sample is compared with the same donor's sham (0 Gy) sample, which removes person-to-person differences. Donor 7 of OSD-1012 has no sham sample and was excluded.
- **Lesson learned:** earlier exports (m3.csv, All.csv) had been cut at exactly 1,048,575 rows, the Excel row limit, which silently removed most sham samples. The cleaned file is never opened in spreadsheet software.

5.4 M4: microbiome swabs (OSD-572)

Skin and oral swab metagenomes of the four Inspiration4 crew were converted to centred log-ratio (CLR) abundances, and each sample's distance from the same person's pre-flight samples was divided by that person's own pre-flight spread (1.0 = typical pre-flight variation).

- **Sites:** body sites were tiered (decisions D11, D12, D14). Tier A ("reliable") keeps sites with low overlap with negative controls and stable pre-flight behaviour; contamination-prone and setting-sensitive sites are excluded from the display.
- **Zero handling:** 58–73% of CLR values are replaced zeros, so confidence is stated as low.
- **Contamination check:** three environmental taxa (Caulobacteraceae, Mesorhizobium, Bradyrhizobium) appear in about 70% of in-flight swabs but only 0–2% of pre-flight swabs. This is likely contamination and is shown as a QC note.

5.5 M6 and M7: Inspiration4 lab panels (OSD-575, OSD-569)

Five processed ("TRANSFORMED") OSDR tables were used, each with the four crew at seven timepoints: three before flight (L-92, L-44, L-3) and four after (R+1, R+45, R+82, R+194). There are no in-flight blood samples.

Table 5. *Inspiration4 laboratory panels used by M7 and M6*

Panel	OSDR study	Analytes	Scale used	Notes
Eve immune 71-plex	OSD-575 serum	71	log2 concentration (pg/mL)	28 samples, no missing values
Alamar NULISA 203-target	OSD-575 serum	203	NPQ (already log2)	27 samples (C003 L-44 missing); zero values treated as missing
Eve cardiovascular	OSD-575 serum	9	log2 concentration	28 samples
CMP (metabolic panel)	OSD-575 serum	17 (16 scored)	raw units	Race-coded eGFR fields removed (creatinine already included); BUN/creatinine ratio is reported for too few samples to score
CBC (blood counts)	OSD-569 whole blood	20	raw units	Long format; 1–2 values missing for some crew

Access and verification. OSDR was not reachable from the build environment, so the files were taken from two public GitHub mirrors of the OSDR downloads and cross-checked: the metabolic panel in both mirrors is identical in all 28 samples × 44 numeric columns; an independent third project reports the same C003 R+1 potassium (3.0 mmol/L) and glucose (103 mg/dL); and the analyte counts match the OSDR descriptions. SHA-256 checksums of every file are stored with the software and checked by a test, so a changed file is noticed immediately.

5.6 M8: telomere study

Telomere length was measured by multiplex qPCR on dried blood spots collected before, during and after flight (Garcia-Medina et al. 2024, Precision Clinical Medicine). Only group-level results are published: telomeres were longer in flight in all four crew and shorter after return in three of four. Because per-crew values are not available (decision D3), M8 uses these published statements as context and draws no per-person chart.

5.7 M9: mission crew telemetry

The mission scenario has four crew members (SIM-A Commander, SIM-B Pilot, SIM-C Mission Specialist 1, SIM-D Mission Specialist 2) with four daily vital signs over 90 mission days (day −45 to day +45; mission day 1 = launch on 15 February 2030, no day 0). Each person has their own level and day-to-day spread;

the series include a heart-rate-variability dip for SIM-A on days 20–23, a resting-heart-rate rise for SIM-B on days 25–26 and a 14-day data gap for SIM-C (days 4–17). The series were reconstructed from the M9 notebook output (its 14 flag cards give each person's median and MAD), and the rule engine reproduces exactly the notebook's 14 flags and no others. Cabin CO2 is recorded daily as context.

5.8 Reference data stored with the software

- **rules_v1.json**: every rule, threshold, action text and the list of forbidden diagnostic words.
- **evidence_registry.json**: 14 evidence entries (datasets, papers, standards) linked to the rules they support.
- **mission.json**: mission timeline, crew roles, storm timing, cabin-CO2 settings.
- **m8_context.json**: the published telomere statements with citation.
- **m1_model_export.json**: M1 coefficients exported from the trained model files.

6. Methodology

This chapter describes the two machine-learning models, the physics estimate and every rule engine, with their validation results. All fixed thresholds are versioned in rules_v1.json; the data-driven thresholds of M7 and M6 are calibrated on pre-flight data only, never on post-flight results.

6.1 Evidence gate: which method for which domain

Table 6. *Evidence gate by domain*

Domain	Human data found	Tier	Module
Solar storm forecasting	DONKI + GOES event history 2010–2026	Tier 1 · ML	M1
Crew dose	MSL/RAD background rate	Estimate	M2
Blood radiation response	OSD-1012 (24 h), OSD-410 (1/2/6 h); OSD-71 external test	Tier 1 · ML	M3
Microbiome	OSD-572 swabs, 4 crew, Tier A sites	Tier 2 · trend	M4
Brain	Mouse studies only	Tier 3 · context	M5
Recovery	OSD-575 + OSD-569 panels to R+194	Tier 2 · trend	M6
Immunology	OSD-575 Eve + NULISA, OSD-569 CBC, 4 crew	Tier 2 · trend	M7
Telomere	Published group results only	Tier 3 · context	M8
Vital signs	Mission telemetry; rule from Iglewicz & Hoaglin 1993	Tier 4 · rules	M9
Bone	Published recovery parameters; scans if entered	Tier 3 · context	Bone panel
Behaviour	Self-report + in-app reaction test	Tier 4 · self-report	Behaviour

6.2 M1: solar particle storm forecaster (machine learning)

Model

- **Algorithm:** logistic regression (scikit-learn pipeline: StandardScaler → LogisticRegression, L2 penalty, class_weight = balanced), followed by Platt calibration (a second logistic regression on the decision value).
- **Feature:** flare_log10_peak_flux (log10 of the GOES X-ray peak flux of the flare).
- **Decision rule:** alert when the calibrated probability ≥ 0.05; in plain terms, a high alert for flares of class M5.0 or larger.
- **Formula:** $z = (x + 4.628) / 0.357$; $d = 1.121\,z - 0.646$; $p = 1 / (1 + \exp(-(0.847\,d - 3.313)))$, where x is the log10 peak flux.

Model selection on training data only

Three candidates were compared with cross-validation inside the training period. The simplest model, flare size alone, had the best precision–recall area, so it was chosen; the richer models ranked storms slightly better but did not find them more precisely.

Table 7. *M1 cross-validation comparison (training period 2010–2023)*

Model	PR-AUC	ROC-AUC	Brier
Baseline (flare size) ✓ chosen	0.241	0.804	0.165
LR (7 features)	0.218	0.847	0.147
LR (6 features)	0.168	0.819	0.165

Results on the untouched test period (2024–2026)

14 / 21	**0.811**	**247**	**0.013**
storms warned (recall 66.7%)	ROC-AUC (ranking skill)	false alerts (≈ 7.3 per month)	Brier score

Table 8. *M1 test-set results*

Metric	Value
Storms warned / missed	14 / 7
Recall (95% CI)	66.7% (44.4–87%)
PR-AUC (95% CI)	0.084 (0.041–0.171)
ROC-AUC	0.811
False alerts	247 (94.6% of alerts)
Brier score	0.013
May 2024 storm (X3.9 flare, 10 May)	warned, p = 0.349

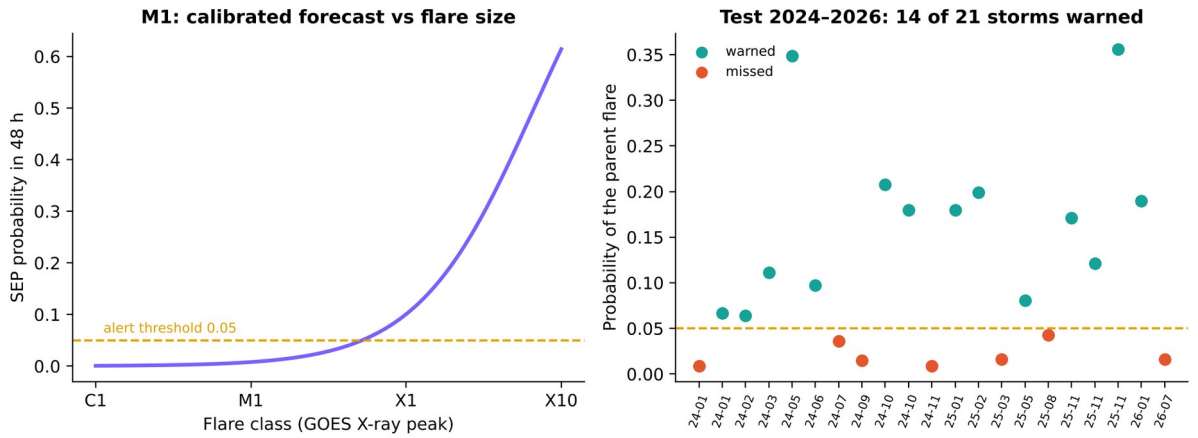


Figure 2. *Left: calibrated SEP probability rises with flare size; the alert threshold corresponds to about an M5 flare. Right: probability given to the parent flare of each of the 21 test storms.*

Limitations of M1

- Few storms (wide intervals).
- Storm rate lower in test than train: probabilities over-stated ~3x on recent data.
- Many false alarms (flare-only forecast of a rare event).
- Validation: the notebook exports reproduce exactly in the software (all 1,720 test probabilities match to 3 decimals).

6.3 M3: blood radiation biodosimeter (machine learning)

M3 estimates radiation exposure from the change in blood gene expression relative to the person's own baseline sample. Four separate models were trained, one per time window, because the genes that respond after one hour are not the same as those after one day.

Table 9. *M3 models (all validated leave-one-donor-out with nested grid search)*

Model	Training data	Task	Features	Hyper-parameters	Validation result
M3-24h-v2	OSD-1012 X-ray, 11 donors, 49 samples	Regression on log2(Gy + 0.1)	11 genes	Elastic-net α 0.01, L1 ratio 0.8; top-100 pre-selection	LODO exposed: MAE 0.379 Gy, 81.6% within 0.5 Gy
M3-early-1h-v2	OSD-410 γ, 10 donors, 40 samples	4 dose classes	50 genes	L1 logistic, C 1; top-50	Balanced accuracy 0.60 vs chance 0.33
M3-early-2h-v2	OSD-410 γ, 10 donors, 40 samples	4 dose classes	50 genes	L1 logistic, C 10; top-50	Balanced accuracy 0.60 vs chance 0.33
M3-early-6h-v2	OSD-410 γ, 5 donors, 20 samples	4 dose classes	20 genes	L1 logistic, C 10; top-20	Balanced accuracy 0.80 vs chance 0.33

Validation design

- **Leave-one-donor-out (LODO):** each donor is held out in turn, and gene selection plus hyper-parameter search happen inside each training fold, so the held-out donor never influences the model.
- **Own-baseline input:** the model sees exposed − baseline for each gene, never absolute expression.
- **Neutron test:** the frozen 24 h model was applied to the OSD-1012 neutron arm (37 samples), which it never saw.
- **External test:** the 1 h model was applied once to OSD-71, an independent lab and platform.

Table 10. *M3-24h leave-one-donor-out error by dose*

True dose (Gy)	n	Mean prediction (Gy)	MAE (Gy)	Bias (Gy)
0	11	0.002	0.002	0.002
0.1	5	0.077	0.046	-0.023
0.3	5	0.237	0.08	-0.063
0.5	5	0.342	0.186	-0.158
1	11	0.882	0.202	-0.118
2	6	1.825	0.3	-0.175
4	6	3.226	1.472	-0.774

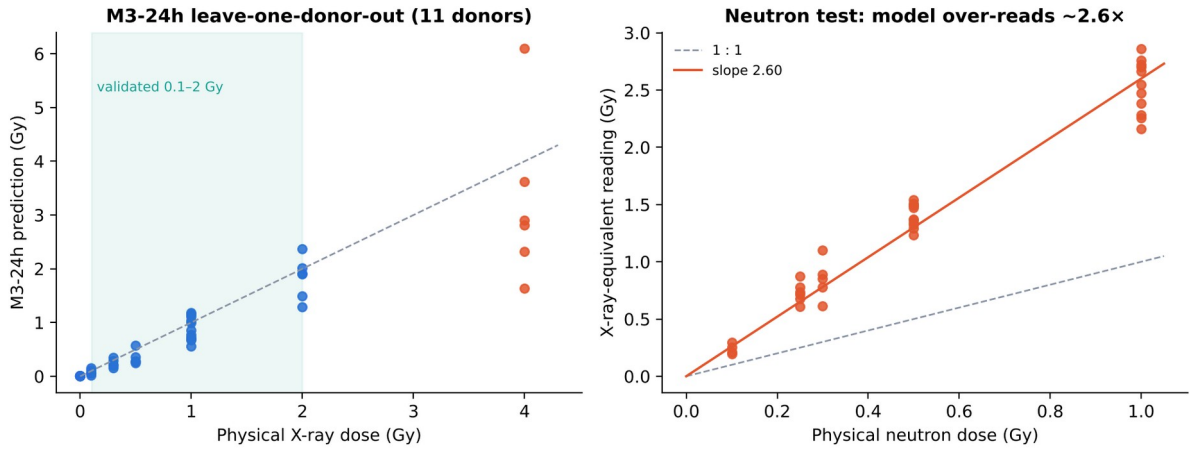


Figure 3. *Left: M3-24h predictions for held-out donors (orange = 4 Gy, under-read). Right: the X-ray-trained model reads neutron dose about 2.6 times higher, so readings are reported as X-ray-equivalent.*

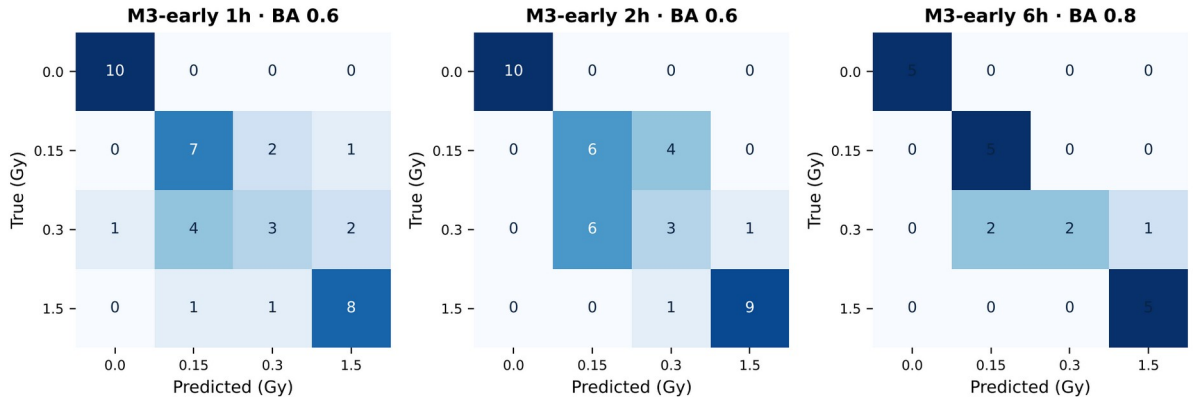


Figure 4. *Confusion matrices of the early models (held-out donors). 1.5 Gy is recognised well; 0.15 and 0.30 Gy are often confused.*

Table 11. *The 11 stable genes of M3-24h (stability selection, ≥ 60% of 100 bootstraps)*

Gene	Probe	Selected in (bootstraps)	Weight
FDXR	A_23_P38154	100%	0.845
ASTN2	A_33_P3371904	97%	0.123
DDB2	A_23_P52610	94%	0.220
GRIN3B	A_33_P3394213	88%	0.033
C3orf52	A_23_P58009	87%	0.130
GADD45A	A_23_P23221	87%	0.119
AEN	A_33_P3214501	84%	0.010
PHLDA3	A_33_P3252834	81%	0.121
BBC3	A_23_P382775	81%	0.126
NCR3	A_23_P251881	72%	-0.202
MDM2	A_24_P925664	64%	0.005

FDXR, DDB2, GADD45A, PHLDA3, BBC3, AEN and MDM2 are known p53-pathway radiation-response genes; ASTN2, GRIN3B and C3orf52 have small weights and are likely spurious.

Key findings and safeguards for M3

- **Valid range:** 0.1–2 Gy, 20–28 h after exposure; 4 Gy is under-read (mean prediction 3.226 Gy).
- **Neutrons are over-read:** slope 2.60, so all readings are labelled X-ray-equivalent.
- **External test failed at 1 h:** on OSD-71 the 1 h model predicted 0 Gy for all 12 exposed samples (balanced accuracy 0.00 vs chance 0.50). The key genes barely respond at 1 h in that lab (1.5 Gy: CDKN1A 1.31 vs 0.08, AEN 1.21 vs 0.17, GADD45A 0.72 vs 0.08 log2 change), so the 1 h model is marked exploratory.
- **Window rule R02:** no estimate unless the time since exposure is 1, 2 or 6 h (±0.5 h) or 20–28 h and the person's own baseline sample exists.
- **Scope:** trained on ex vivo irradiated Earth blood, not astronauts.

6.4 M2: crew dose estimate (physics)

Daily dose-equivalent = 1.8 mSv/day galactic cosmic-ray background + k × (mean ≥10 MeV proton flux above background, pfu) × shielding factor, with k = 0.0006 mSv/day per pfu. The location on each day sets the shielding factor; crew member SIM-D is on EVA on the storm onset day. Cumulative mission dose is also shown as a share of NASA's 600 mSv career limit (NASA-STD-3001). Over the 45 in-flight days the estimate is 83.22 mSv for SIM-A (storm adds 2.16 mSv) and 85.54 mSv for SIM-D (storm adds 4.48 mSv, because of the EVA). The storm coefficient and the shielding factors are set for this mission scenario and should be calibrated against MSL/RAD storm measurements.

6.5 Rule-based monitoring

M9-FLAG-v1, M9-MULTI-v1, M9-Q01: daily vital signs

For each crew member, vital sign and day, the previous 30 days form the personal baseline (at least 21 valid days are required). The modified z-score of Iglewicz and Hoaglin (1993) is used because it is robust to outliers:

```
modified z = 0.6745 × (today − median of previous 30 days) ÷ MAD
flag when |modified z| ≥ 3.5 and at least 21 valid baseline days exist
```

Table 12. *Vital-sign rules*

Rule	Fires when	Result
M9-FLAG-v1	|modified z| ≥ 3.5 vs own previous 30 days	Flag + explanation card; alert for HR, HRV, respiratory rate; SpO2 flagged but never alerts (readings vary too little for the statistic to be reliable)
M9-MULTI-v1	2 or more alert-enabled vitals flagged on the same day	Escalated recommendation (R08)
M9-Q01	Fewer than 21 valid days in the previous 30, or no reading today	Rules paused; "collect another measurement" (R03)

Over the mission the rules raise **14 flags** (7 alerts on heart rate, HRV and respiratory rate; 7 SpO2 flags without alert), identical to the M9 notebook output. Figure 5 shows the HRV dip of SIM-A on days 20–21.

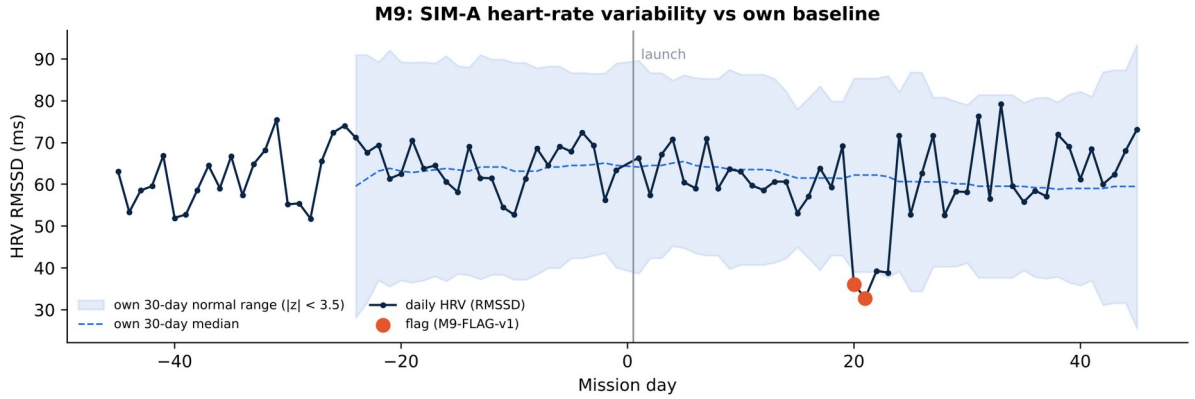


Figure 5. *M9 for SIM-A: daily HRV, own 30-day median and normal range; the rule flags days 20 and 21.*

M4-DRIFT-v1: microbiome

Status = "within own pre-flight range" when the median in-flight distance (divided by own pre-flight spread) lies inside the person's own pre-flight range; Tier A sites only. All four crew are within their own range in flight and back near baseline by R+82:

Table 13. *M4 results (Inspiration4, OSD-572)*

Crew	In-flight distance / spread	Own pre-flight range	R+82	Status	Tier A sites
C001	1.14	0.71 – 1.38	0.86	Within own pre-flight range	Gluteal Crease, Oral, Toe Web Space, Umbilicus
C002	0.86	0.79 – 1.27	0.88	Within own pre-flight range	Axillary Vault, Oral, Toe Web Space
C003	1.1	0.66 – 1.36	0.97	Within own pre-flight range	Axillary Vault, Gluteal Crease, Oral
C004	1.02	0.77 – 1.31	0.88	Within own pre-flight range	Axillary Vault, Gluteal Crease, Oral, Toe Web Space, Umbilicus

An exploratory classifier (pre-flight vs in-flight, leave-one-crew-out, 97 samples) reached AUROC 0.987 on species CLR, but dropped to 0.632 (chance 95th percentile 0.628) when the top 10 in-flight-only taxa were removed. The apparent signal came from a few environmental taxa, so it is **not used as a prediction model**.

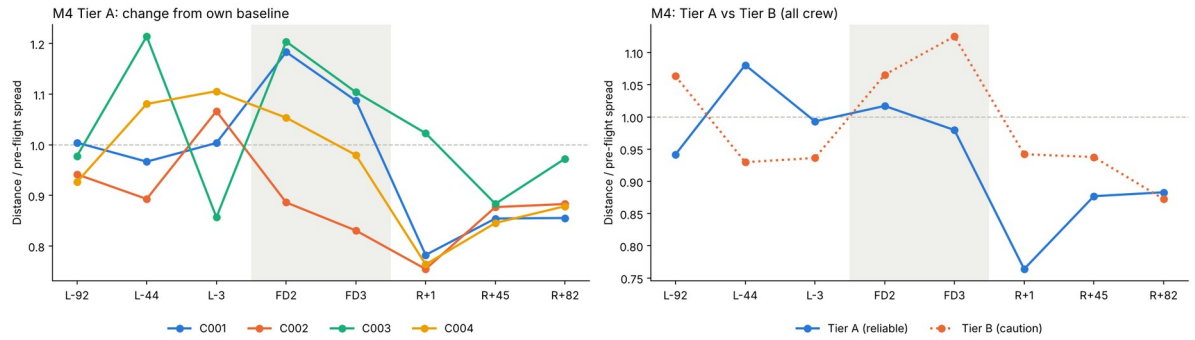


Figure 6. *M4 Tier A distance from own baseline per crew member, and Tier A vs Tier B (team notebook).*

M7-DEV-v1: immune and inflammatory monitoring

M7 is computed inside the software from the OSD-575 and OSD-569 panels, separately for each panel:

- **Baseline:** the mean of the person's own pre-flight samples (L-92, L-44, L-3).
- **Spread:** the pooled within-person pre-flight standard deviation of each analyte across the four crew.
- **Score:** z = (value − own baseline mean) ÷ (pooled SD × √(1 + 1/n_baseline)).
- **Threshold k, calibrated on pre-flight data only:** each pre-flight sample is left out in turn and scored against the person's other pre-flight samples; k is the 95th percentile of these |z| values, so about 5% of pre-flight samples would be flagged.
- **Chance test:** for each crew member and timepoint, the number of flagged analytes is compared with the 5% expected by chance (one-sided binomial test, p < 0.05 = "above chance").

Table 14. *M7 thresholds per panel (calibration works as designed: 5.0–5.6%)*

Panel	Analytes scored	Threshold k	Measured pre-flight false-flag rate
Serum cytokines (Eve 71-plex)	71	2.736	5.1%
Serum inflammation (Alamar NULISA 203-target)	203	2.431	5.0%
Blood counts (CBC)	20	3.147	5.1%
Serum cardiovascular (Eve)	9	2.193	5.6%
Metabolic panel (CMP)	16	2.672	5.2%

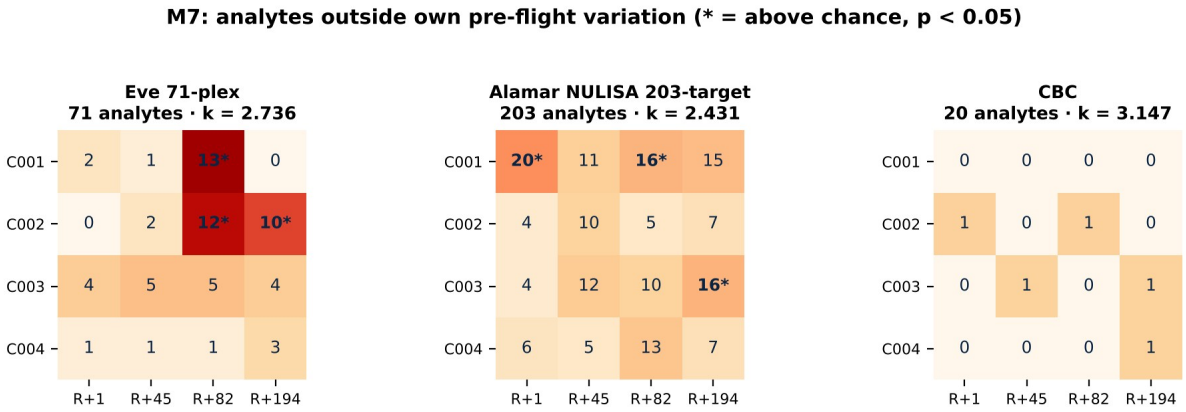


Figure 7. *Number of analytes outside own pre-flight variation per crew member and timepoint (* = above chance).*

Table 15. *All crew × timepoint results above chance*

Panel	Crew	Timepoint	Flagged / tested	Expected	p
Eve 71-plex cytokines	C001	R+82	13 / 71	3.6	< 0.0001
Eve 71-plex cytokines	C002	R+82	12 / 71	3.6	0.0002
Eve 71-plex cytokines	C002	R+194	10 / 71	3.6	0.0027
Alamar NULISA 203-target	C001	R+1	20 / 203	10.2	0.0032
Alamar NULISA 203-target	C001	R+82	16 / 203	10.2	0.0496
Alamar NULISA 203-target	C003	R+194	16 / 202	10.1	0.0478
Eve cardiovascular	C002	R+1	3 / 9	0.5	0.0084

Panel	Crew	Timepoint	Flagged / tested	Expected	p
CMP	C003	R+45	3 / 16	0.8	0.0429

Agreement with the team notebook

• **Notebook:** only C001 and C002, Eve panel, R+82 exceed chance (11/71 vs 3.6 expected).

• **Software:** Eve panel R+82 above chance for C001 (13/71) and C002 (12/71) and not for C003 or C004, the same pattern.

• Additional above-chance cells appear in the NULISA, cardiovascular and metabolic panels, several of them borderline (p ≈ 0.04–0.05).

• Because analytes are correlated and many comparisons are made (4 crew × 4 timepoints × several panels), "above chance" is descriptive, not a significance claim.

M6-TRAJ-v2: recovery after landing

Every analyte of the five panels that is outside its own pre-flight variation at R+1 (same threshold k as M7) is followed to R+45, R+82 and R+194; it counts as "back" when |z| ≤ k. Across the four crew, **46 analytes changed at R+1** and **34 of 46 (74%) were back within own variation by R+82** (38 by R+45, 31 by R+194). The team notebook reported 35 of 39 with its own thresholds.

Table 16. *M6 recovery per crew member*

Crew	Changed at R+1	Back by R+45	Back by R+82	Back by R+194
C001	22	18	13	13
C002	9	9	8	7
C003	8	4	7	5
C004	7	7	6	6
All	46	38	34	31

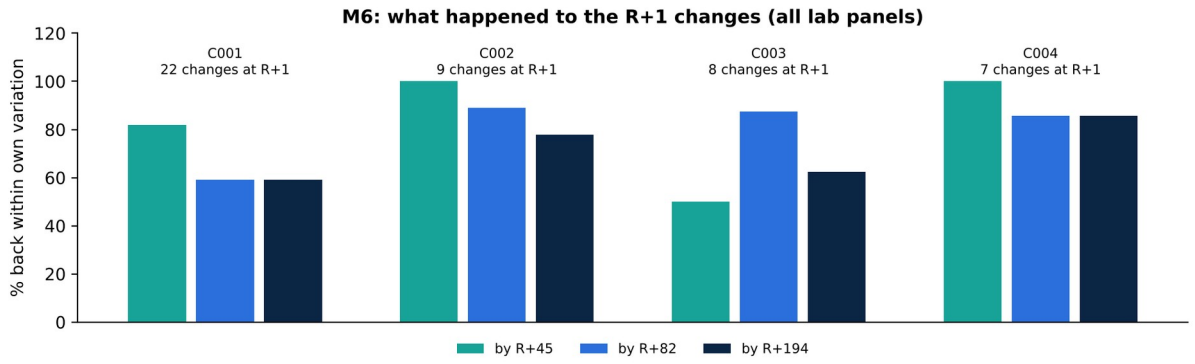


Figure 8. *Share of R+1 changes back within own variation at each later timepoint.*

Action engine: rule-based recommendations

Rules run in priority order: 1 = data-quality and validity rules first, 2 = status rules, 3 = routine. Every recommendation shows its rule ID and the reason. A test checks every recommendation for forbidden words (diagnosis, disease, medication, treatment and similar).

Table 17. *Action rules (rules_v1)*

Rule	Priority	Name	Fires when	Recommendation
R02	1	Outside validated window	M3 request outside 1 h / 2 h / 6 h / 20-28 h, or no own baseline sample	Outside validated analysis window. Do not interpret a model output.
R03	1	Insufficient data	M9-Q01 active for a measure	Insufficient evidence for reliable interpretation. Collect another measurement.
R09	2	Vitals changed from own baseline	M9-FLAG-v1 fires on an alert-enabled measure	Check the sensor and repeat the measurement; review with flight surgeon if it persists.
R08	2	Several vitals changed	M9-MULTI-v1 fires	Repeat all measurements and review with the flight surgeon.
R10	2	Radiation event	M1 alert in the last 48 h, or M2 dose rate above background	Review radiation exposure, follow the storm-shelter protocol, and continue scheduled monitoring.
R11	2	Blood response elevated	M3 reading above 0 Gy class / >= 0.1 Gy inside a validated window	Seek medical review: review the blood-response result with the flight surgeon.
R12	2	Physical and biological dose disagree	M2 estimate low but M3 elevated, or the reverse	Repeat the blood assessment at the next validated window.
R20	2	Immune change	M7-DEV-v1: a panel has more analytes outside own pre-flight variation than expected by chance (binomial $p < 0.05$)	Continue immune monitoring and review the mission medical protocol.
R30	3	Microbiome monitoring	M4 card available	Continue longitudinal microbiome monitoring.
R40	3	Cellular-aging monitoring	M8 published-context card	Continue longitudinal telomere monitoring and compare with recovery measurements.
R50	2	Behaviour changed	2 or more self-report items worse than own first 7 entries	Repeat the self-assessment and review the mission behavioural-health support procedures.
R60	2	No bone measurement	more than 30 days in microgravity and no scan entered	Obtain a bone measurement per mission protocol.
R70	3	Recovery monitoring	M6-TRAJ-v2 result available (post-flight replay)	Continue longitudinal recovery monitoring.
R99	3	Routine	no other status rule active	Continue scheduled monitoring.

Provenance: seven status types

Table 18. *Status types shown on every value*

Status type	Meaning	Examples
MEASURED	Directly measured	Bone scan entry, reaction time, Inspiration4 blood counts
DERIVED	Computed from measured values	M4 distance, M7/M6 flags, explanation deviation
ML_PREDICTED	Trained model output	M1 SEP probability, M3 dose
ESTIMATED	Physics or approximation	M2 dose rate and mission dose
CONTEXT	Timeline or published parameter	Mission phase, M5, M8, bone context
SELF_REPORTED	Entered by the astronaut	Daily self-check
TELEMETRY	Crew vitals and cabin telemetry stream	M9 vital signs, cabin CO2

6.6 Testing and verification

The software ships with 26 automated tests (python -m unittest discover -s backend/tests). They check that:

- M1 reproduces all 1,720 notebook test probabilities and the 14 caught / 7 missed storm counts;
- the M3 port gives exactly the same output as the team's original numpy predictor, and refuses outside validated windows, without a baseline or with missing genes;
- M9 reproduces exactly the notebook's 14 flags, deterministically, with SpO2 never alerting and no flag without 21 baseline days;
- the Inspiration4 data files are unchanged (checksums), complete for all crew and timepoints, free of race-coded fields, and NULISA zeros are treated as missing;
- M7 thresholds give 5% ± 1% pre-flight false flags and the Eve R+82 result agrees with the notebook; M6 counts are consistent; M8 is context only;
- every recommendation has a rule ID and reason and no forbidden wording; every displayed value has a known status type; every API route responds and bad input is rejected.

In addition, every page was opened for every crew member in a headless browser to check for script errors and empty or broken values.

7. Software Functions

The console is a single web application served by the local backend. A crew selector at the top right switches between the mission crew and the Inspiration4 crew depending on the page. This chapter describes every page and its functions.

7.1 Home

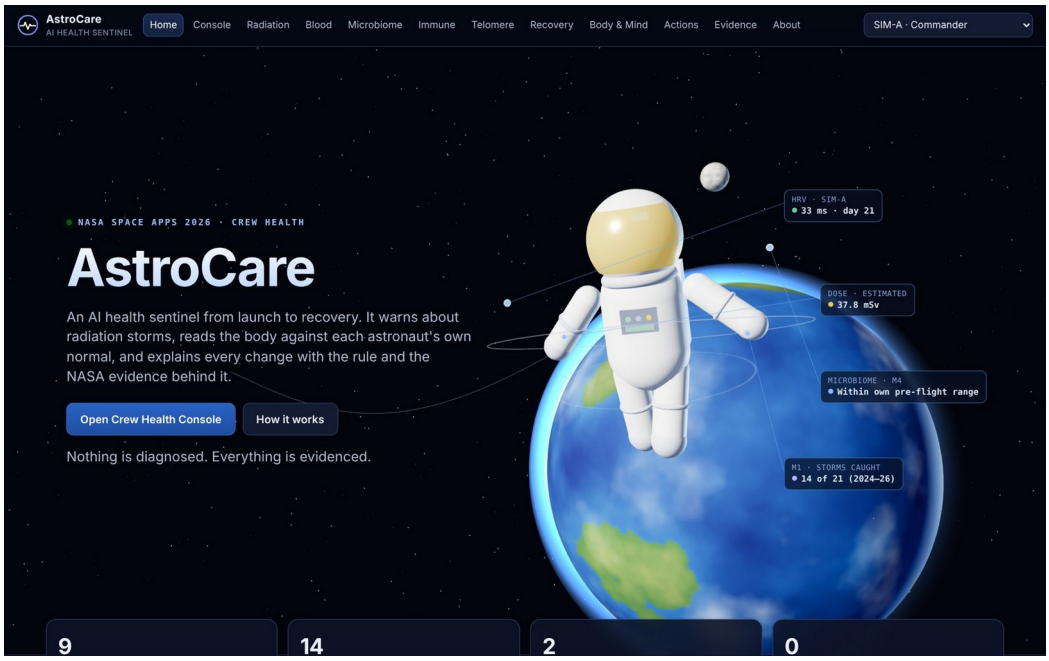


Figure 9. *Home page.*

- 3D scene of an astronaut over Earth with four live indicators (HRV, mission dose, microbiome status, M1 storms warned).
- Nine module cards that open the matching page, and the six-step pipeline from data to action.

7.2 Crew Health Console (M9)

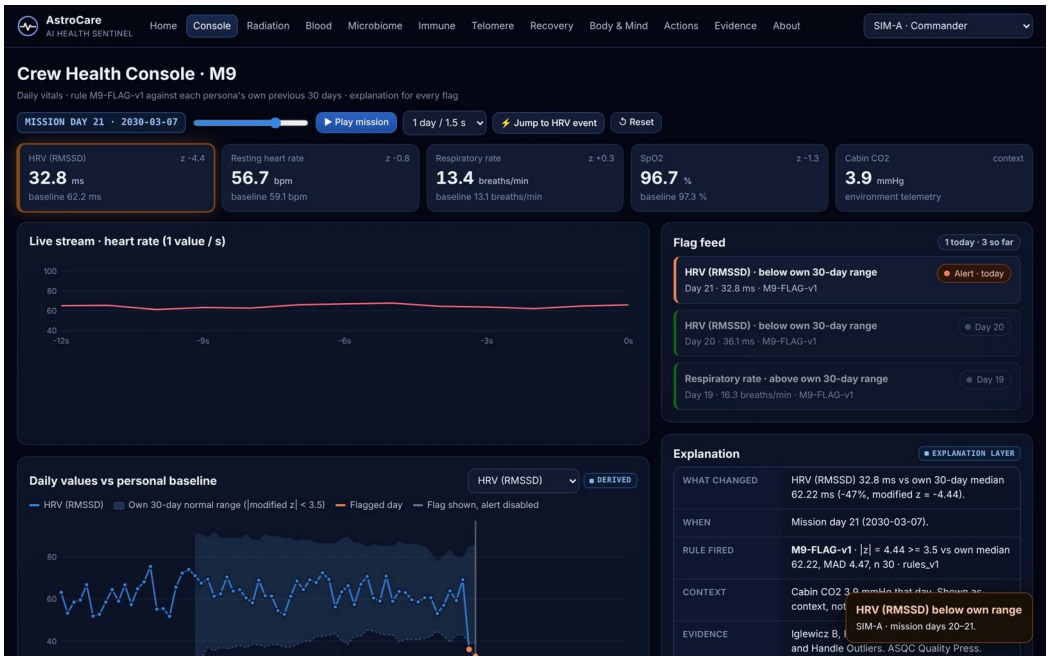


Figure 10. *Crew Health Console at the SIM-A HRV event (mission day 21).*

- Mission-day control: slider, ▶️ Play mission (1 day per 3, 1.5 or 0.6 s), ⚡ Jump to HRV event, ↺ Reset.
- Tiles for HRV, resting heart rate, respiratory rate, SpO2 (value, modified z, own baseline) and cabin CO2.
- Live heart-rate stream (one value per second).
- Daily chart of the selected vital with the own 30-day normal range, flagged days and cabin CO2 underneath.
- Flag feed (alerts, past flags, SpO2 flags without alert, data-quality pauses) and an Explanation card: what changed, when, rule fired, context, evidence, "not a diagnosis", status type and recommended action.
- Rule table of M9-FLAG-v1, M9-MULTI-v1 and M9-Q01.

7.3 Radiation (M1 + M2)

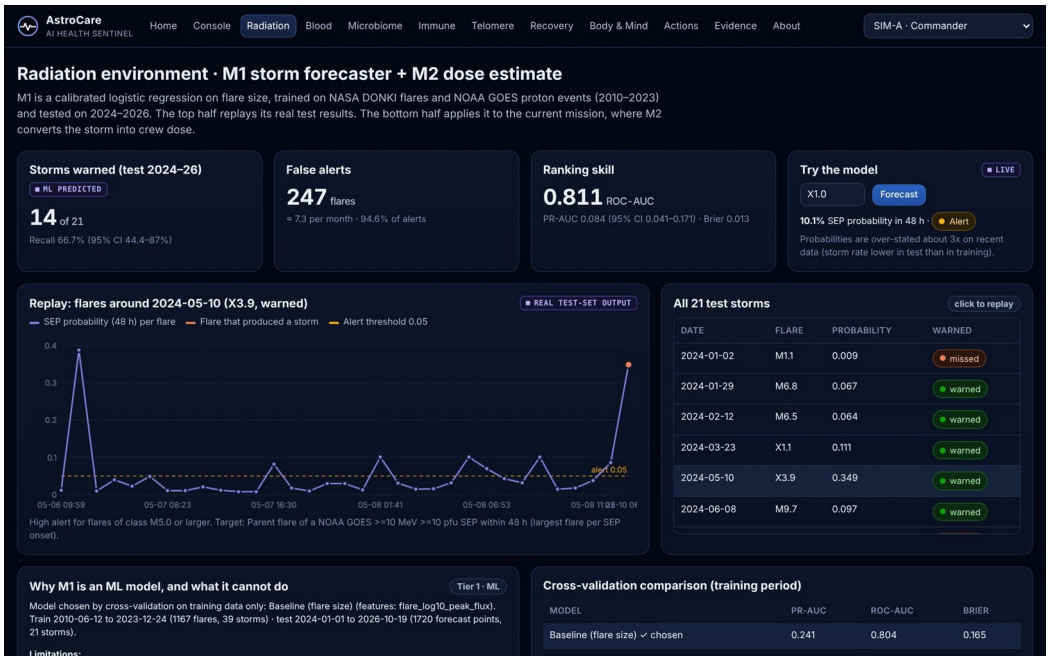


Figure 11. *Radiation page.*

- Real M1 test results: storms warned, false alerts, ROC-AUC, and a "Try the model" box that forecasts any flare class (for example X1.0).
- Replay of the flares around any of the 21 test storms, with the alert threshold; table of all storms (warned or missed).
- Model-selection table and limitations.
- Current mission: SEP probability, dose rate, cumulative dose (% of the 600 mSv limit), status, "go to storm day" and a location selector (habitat, shelter, EVA); proton-flux and cumulative-dose charts for all crew.

7.4 Blood (M3)

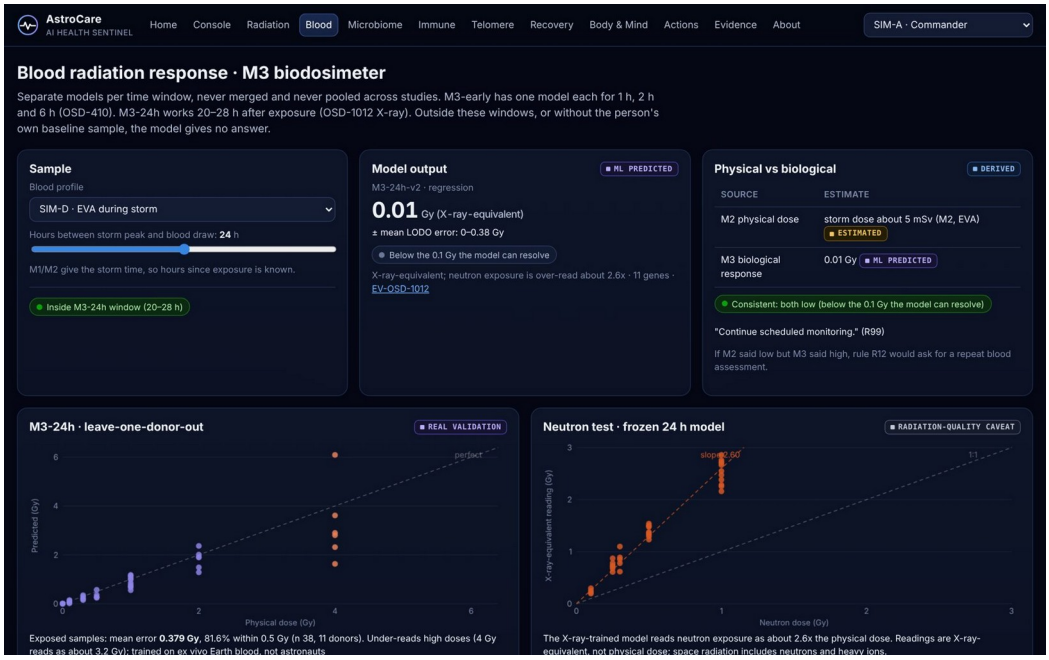


Figure 12. *Blood page.*

- Choose a blood profile and the hours between storm peak and blood draw; the window indicator shows which model is valid.
- Model output: dose class with probabilities (1, 2, 6 h) or X-ray-equivalent dose with error band and range check (20–28 h); "No output" outside validated windows.
- Physical (M2) vs biological (M3) comparison with rule R11 / R12 / R99.
- Validation section: LODO scatter, neutron test, three confusion matrices, gene table, OSD-71 external check.
- Run the model on your own sample: paste or upload JSON with exposed and baseline expression.

7.5 Microbiome (M4)

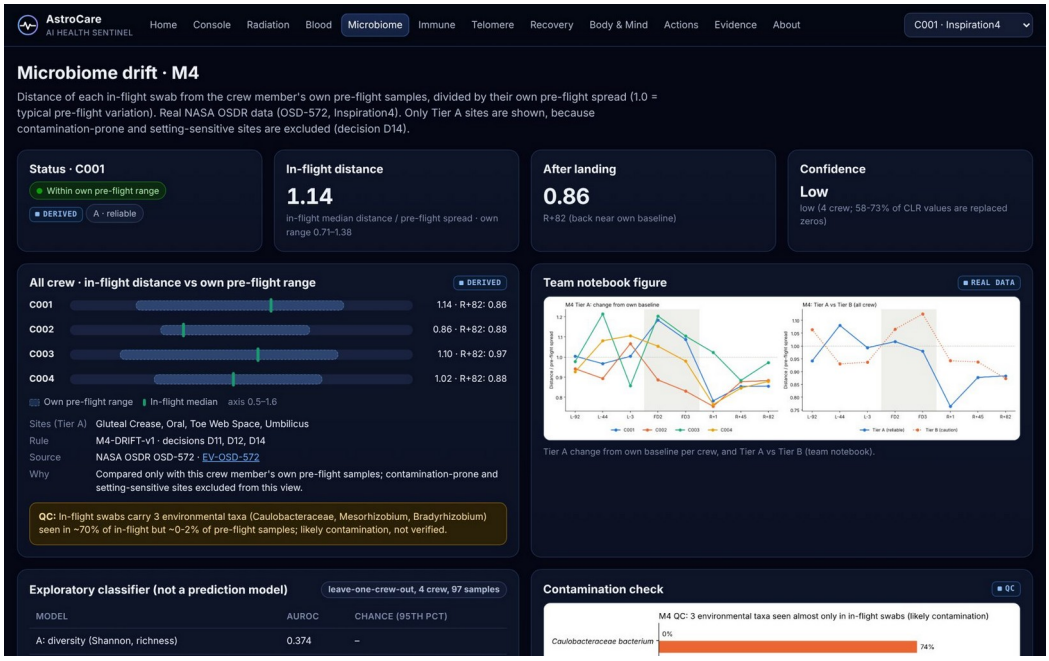


Figure 13. *Microbiome page (C001).*

- Status, in-flight distance, after-landing value and confidence for the selected crew member.
- All-crew range bars, Tier A sites, QC note, team figures, exploratory-classifier table and contamination check.

7.6 Immune (M7)

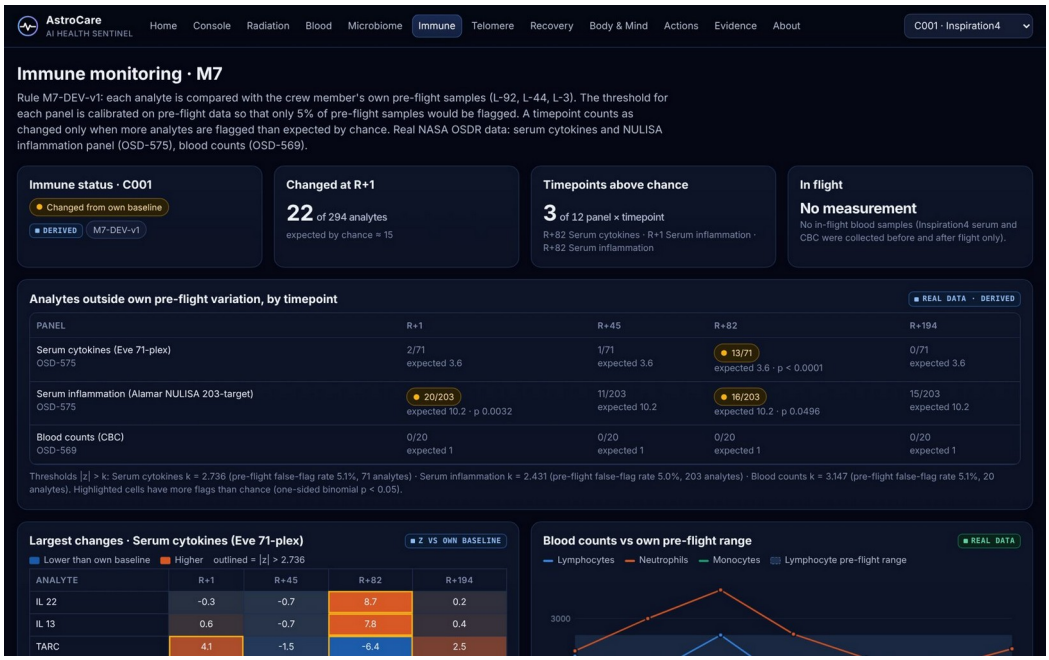


Figure 14. *Immune page (C001).*

- Immune status, number of analytes changed at R+1, timepoints above chance, and an in-flight "no measurement" card.
- Table of flagged / tested / expected / p per panel and timepoint; heat-map of the largest changes (z vs own baseline); blood counts vs own pre-flight range.

7.7 Telomere (M8)

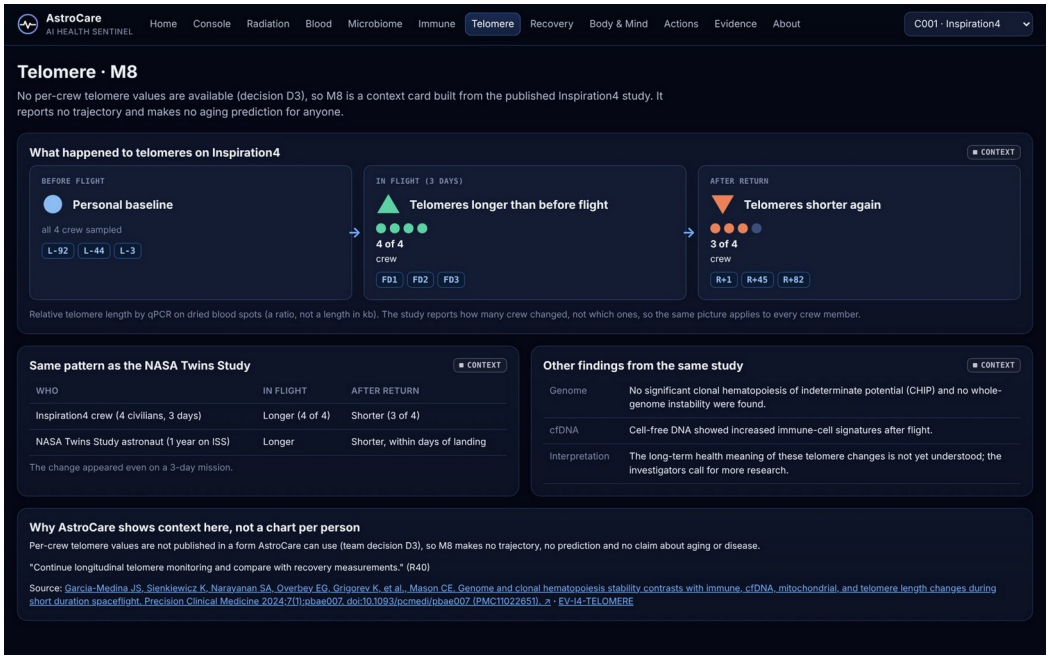


Figure 15. *Telomere page.*

- Visual timeline: before flight (baseline) → in flight (longer in 4 of 4 crew) → after return (shorter in 3 of 4).
- Comparison with the NASA Twins Study, other findings of the same study, and why the page shows context rather than a per-person chart.

7.8 Recovery (M6)

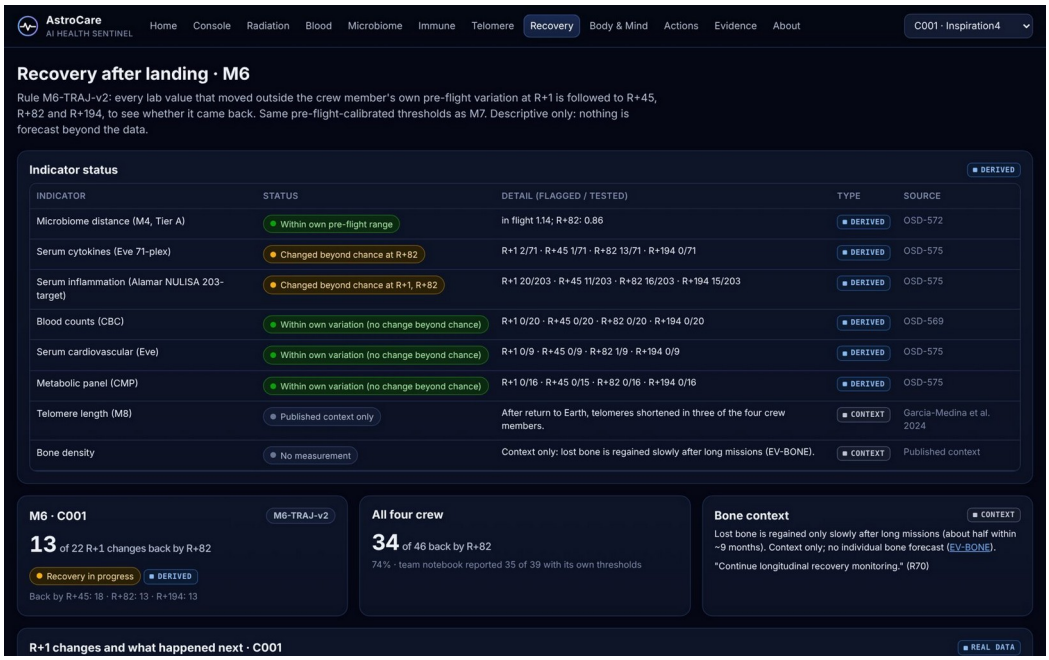


Figure 16. *Recovery page (C001).*

- Indicator status table for microbiome, five lab panels, telomere and bone.
- R+1 changes back by R+82 for the crew member and for all four crew; table of every R+1 change and what happened next.

7.9 Body & Mind

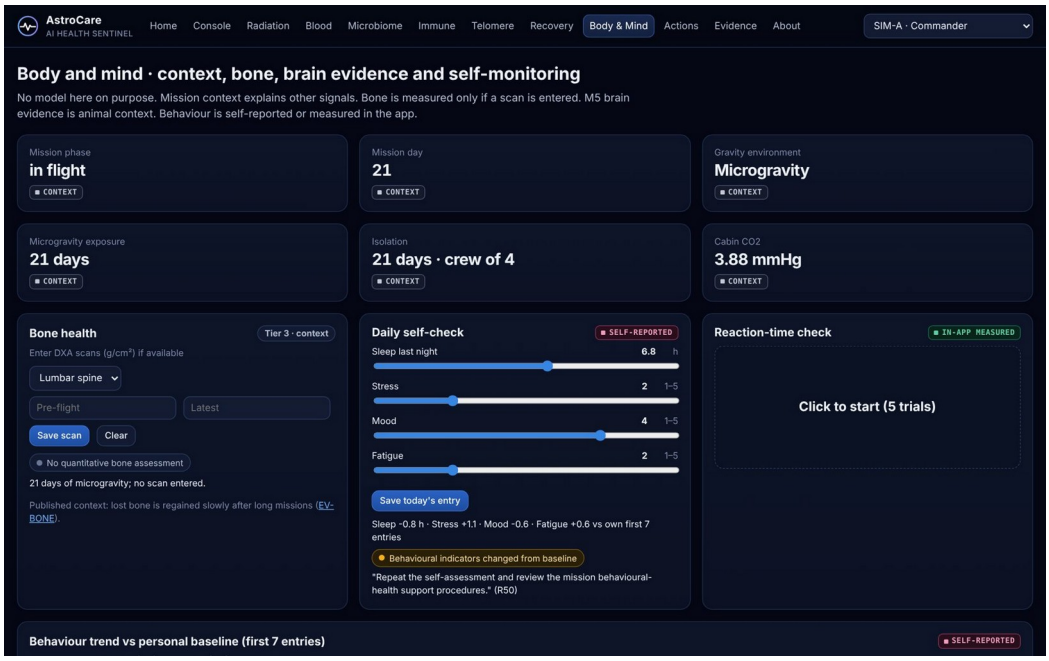


Figure 17. *Body & Mind page.*

- Mission context tiles: phase, mission day, gravity, microgravity exposure, isolation, cabin CO2.
- Bone panel: enter pre-flight and latest DXA scans to see the % change (R60 asks for a scan after 30 days).
- Daily self-check sliders (sleep, stress, mood, fatigue) saved to the backend, with rule R50 and trend charts against the first 7 entries.

- Five-trial reaction-time test saved and compared with the person's first three runs; M5 brain evidence card.

7.10 Actions

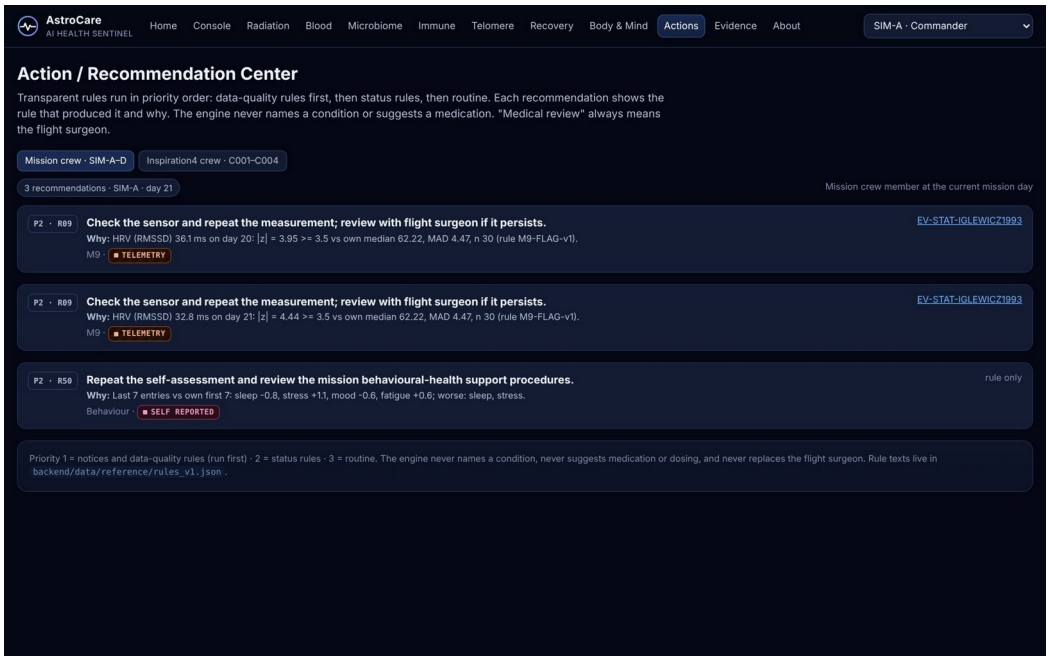


Figure 18. *Action / Recommendation Center.*

- All recommendations for the selected crew member (and mission day), sorted by priority, each with rule ID, reason, module, status type and evidence link.
- Switch between mission crew and Inspiration4 crew.

7.11 Evidence and About

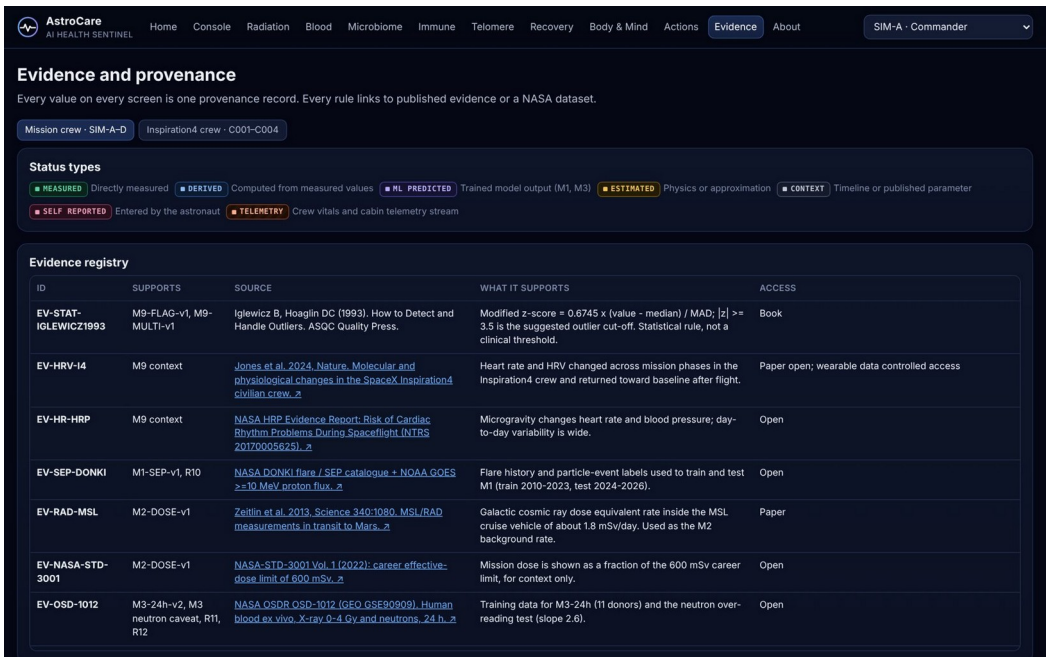


Figure 19. *Evidence page.*

- Evidence page: the seven status types, the evidence registry (14 entries with links) and every provenance record for the selected subject.

- About page: architecture, evidence gate by domain and link to the interactive API documentation (/docs).

7.12 Backend API

All pages read from the local backend; 33 endpoints are available (interactive documentation at /docs when FastAPI is installed).

Table 19. *API endpoints*

Method	Endpoint	Method	Endpoint
GET	/api/health	GET	/api/immune/{crew_id}
GET	/api/meta	GET	/api/telomere/{crew_id}
GET	/api/crew	GET	/api/recovery/{crew_id}
GET	/api/mission/{crew_id}/context	GET	/api/brain
GET	/api/vitals/rules	GET	/api/i4/calibration
GET	/api/vitals/{crew_id}	GET	/api/selfreport/{crew_id}
GET	/api/vitals/{crew_id}/flags	POST	/api/selfreport/{crew_id}
GET	/api/vitals/{crew_id}/live	POST	/api/reaction/{crew_id}
GET	/api/radiation/m1/summary	GET	/api/bone/{crew_id}
GET	/api/radiation/m1/storms	POST	/api/bone/{crew_id}
GET	/api/radiation/m1/timeline	GET	/api/actions/{subject_id}
POST	/api/radiation/m1/predict	GET	/api/indicators/{subject_id}
GET	/api/radiation/forecast	GET	/api/evidence
GET	/api/radiation/{crew_id}/dose	GET	/api/evidence/{evidence_id}
GET	/api/blood/info	GET	/api/rules
POST	/api/blood/predict	GET	/api/about
GET	/api/microbiome/{crew_id}		

7.13 How to run

1. Install Python 3.9 or newer.
2. Open the astrocare folder in a terminal (or VS Code).
3. Optional, once: python -m venv .venv, activate it, then pip install -r requirements.txt (FastAPI and API docs).
4. Start: python run.py, then open http://127.0.0.1:8000 (stop with Ctrl + C).
5. Tests: python -m unittest discover -s backend/tests -v.

8. Conclusion

AstroCare shows that an astronaut health console can be both useful and honest. It brings nine health domains into one place and treats each according to the evidence behind it. Where NASA data are rich enough, it predicts: M1 warned of 14 of the 21 solar particle storms of 2024–2026 from flare size alone, including the May 2024 storm, and M3 reads X-ray-equivalent dose from blood with a mean error of 0.38 Gy in held-out donors. Where data cover only four people, it compares each crew member with their own normal using rules whose thresholds are calibrated on pre-flight data: M7 found the same R+82 cytokine changes in C001 and C002 as the team notebook, M6 showed that about three quarters of the changes seen the day after landing were back within each person's own variation by R+82, and M4 found no in-flight microbiome change beyond pre-flight variation once contamination-prone sites were excluded. Where only published results exist, as for telomeres, it shows context and makes no per-person claim.

Every value carries its provenance, every flag its explanation, and every recommendation its rule, without naming a disease or a medication. The whole system runs offline from one command, so it could work where Earth is minutes away.

8.1 Limitations

- Only four Inspiration4 crew and three pre-flight samples per person; results are descriptive.
- M1 has few storms (wide confidence intervals) and many false alerts; probabilities are over-stated about three times on recent data.
- M3 is trained on ex vivo irradiated Earth blood with 5–11 donors per model; the 1 h signal was not reproduced in OSD-71; neutron readings are X-ray-equivalent.
- M2's storm coefficient and shielding factors still need calibration against MSL/RAD storm data.
- The urine inflammation panel (OSD-656) is not included; the Inspiration4 lab files were obtained from public mirrors and cross-checked rather than downloaded directly from OSDR.
- Mission telemetry for SIM-A to SIM-D is a mission scenario reconstructed from the M9 notebook, not data from a flown crew.

8.2 Future work

- Calibrate M2 against MSL/RAD measurements during real solar particle events.
- Add the OSD-656 urine panel to M7 and connect real wearable streams to M9.
- Validate M3 on additional independent datasets and on in-flight blood if it becomes available.
- Test the console with flight-surgeon users and extend M1 with real-time DONKI and GOES feeds when a link to Earth exists.

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